

Programming Exam 4

Date/Time: 2026.04.07 08:00 – 08:50

(程式檔命名: 學號_pe4.py, 上傳至 Moodle PE4 上傳區)

Problem: Reverse Column Zigzag Multiplication Table

Write a Python program that fills a 9×9 grid using the standard 9×9 multiplication table in a **zigzag column-wise pattern from right to left**, and prints each multiplication expression as it is placed.

Rules

- Generate all multiplication expressions from **1*1** to **9*9** in the usual order:
 - 1*1, 1*2, ..., 1*9, 2*1, 2*2, ..., 9*9
- Fill the 9×9 grid **column by column**, but start from the **rightmost column** and move toward the **leftmost column**.
- The filling direction alternates by column:
 - The **rightmost column** is filled **top to bottom**
 - The next column is filled **bottom to top**
 - The next column is filled **top to bottom**
 - and so on, until all 9 columns are filled
- Each time an expression is placed into the grid, immediately print it in the format **a*b=c**, padded to width 8 and left-aligned.
- After filling the entire grid, print the final 9×9 grid.

Notes

- There is **no input**.
- Use width 8 for every printed cell in the final grid so that the columns line up neatly.

Sample Output

(以下是你程式執行後須印出的結果)

```
c:\workspace>python pe4.py
Reverse Column Zigzag Multiplication Table:
9*1=9  8*9=72  7*1=7   6*9=54  5*1=5   4*9=36  3*1=3   2*9=18  1*1=1
9*2=18  8*8=64  7*2=14  6*8=48  5*2=10  4*8=32  3*2=6   2*8=16  1*2=2
9*3=27  8*7=56  7*3=21  6*7=42  5*3=15  4*7=28  3*3=9   2*7=14  1*3=3
9*4=36  8*6=48  7*4=28  6*6=36  5*4=20  4*6=24  3*4=12  2*6=12  1*4=4
9*5=45  8*5=40  7*5=35  6*5=30  5*5=25  4*5=20  3*5=15  2*5=10  1*5=5
9*6=54  8*4=32  7*6=42  6*4=24  5*6=30  4*4=16  3*6=18  2*4=8   1*6=6
9*7=63  8*3=24  7*7=49  6*3=18  5*7=35  4*3=12  3*7=21  2*3=6   1*7=7
9*8=72  8*2=16  7*8=56  6*2=12  5*8=40  4*2=8   3*8=24  2*2=4   1*8=8
9*9=81  8*1=8   7*9=63  6*1=6   5*9=45  4*1=4   3*9=27  2*1=2   1*9=9

c:\workspace>
```

(繳交是交 pe4.py, 不是交截圖)

Note: You need to write comments (註解) for each part in your code.